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PENELITIAN KONSERVASI KEANEKARAGAMAN HAYATI DI PT ARUTMIN INDONESIA TAMBANG KINTAP

Jurnal sequestion of carbon and oxygen production

Jurnal mapping of subtrate

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Sequestration of Carbon and Oxygen Production in Upper Stands in The Arboretum of PT Arutmin Indonesia NPLCT

Muriyana Muhana[1], Yogi Swara Putra Mendra[2], Bahtiar[3]
Faculty of Forestry, Universitas Lambung Mangkurat, Indonesia[1]
PT. Arutmin Indonesia Nplct - Tanjung Pemancingan, Kotabaru, Indonesia[2,3]
Email: muriyanamuhana11@gmail.com

Keywords
Biomass, Carbon stocks, Carbon sequestration, Oxygen, Arboretum.

ABSTRACT

This study aims to determine biomass, carbon, carbon uptake, and oxygen production in the arboretum of PT. Arutmin Indonesia-NPLCT. The methods used are purposive sampling and allometrics. The plot size useda is 20 m × 20 m with 12 plots measured. The area of the arboretum is 5.05 ha. Carbon stock measurement is done by multiplying biomass by a conversion rate of 0.47 (47%) while carbon dioxide (CO2) sequestration is calculated by multiplying the average annual biomass growth by the conversion rate of 1.4667 obtained from the photosynthesis equation and oxygen production is measured based on conversion from carbon yield. Based on the results of the study, it showed that the biomass content of stands in the arboretum was 1,067.32 tons, carbon stocks were 501.64 tons, carbon dioxide (CO2) absorption was 1,565.45 tons, and oxygen production was 1,337.71 tons.

INTRODUCTION

Indonesia's forests are the richest and most diverse forests on earth. The Indonesian government owns more than 120 million hectares of secret forest in the country, almost twice the size of France. Of the 4,444,120 million hectares, Indonesia's forests are divided into three categories. This

METHODS

The research was conducted at the Arboretum of PT. Arutmin Indonesia-NPLCT,Jl. Berangas Km. 5,5 Tanjung Pemancingan Sarang Tiung village, Kotabaru Regency, South Kalimantan. The time required to carry out this research is 2 months.

The tools used are GPS (Global Positioning System), cameras for documentation tools, meters to measure distances, measuring tapes to measure diameters. The object in this study, is the stand on the tree contained in the Arboretum of PT. Arutmin-NPLCT.Tree Measurement at the study site. The data collected or obtained from the field is diameter. The biomass formula used is:

Top Biomass Measurement

Table 1. Biomass Calculation Formula

Types of Vegetation	Formula	source
Acacia	$Y=0{,}060255\times D^{2{,}39}$	Purwitasari (2011)
Tropics Common (Branching Tree)	$Y=0{,}11\times\rho\times D^{2{,}62}$	Ketterings et,al., (2001)
Palm <1,3 m	Y = 0,0976 H+0,0706	ICRAF (2009)
Palm >1,3 m	Y = 0,002382 × D 2,3385 × H0,9411	Lubis (2011)

Information:
Y = Biomass (kg)
H = Tree Height (m)
D = Diameter at Chest Level (dbh) (cm)
ρ = Wood Density/Specific Gravity (mg/m^3, kg/m^3, atau g/m^3)

Information:
Bb = Bottom Biomass (kg)
Ba = Top Biomass (kg)
0,25 = Constant Value (IPCC,2006)

Carbon (CO_2) From Biomass

Calculation by Formula (IPCC, 2006) is:

$$C = \text{Biomass} \times \%\text{C-Organik}$$

Information:
C = Carbon from Biomass (kg)
B = bioresource Total (kg)
%C Organik = 0,47% (SNI 7724, 2011)

Reserve Carbon for Ha for Plot

The carbon value per ha for data on the soil surface is:

$$Cn = \frac{\text{Cx}}{1000} \times \frac{10.000}{Lplot}$$

Information:
Cn = Carbon pool on plot (ton/ha).
Cx = Carbon pool on plot (kg).
Lplot = Area of plots in each carbon pool (m2) (SNI 7724, 2011)

Sequestration Karbon (CO_2)

Sequestration CO_2, rumus ialah:

$$\text{Sequestration } CO_2 = \text{Biomass (kg)} \times 1{,}4667$$

Information
1,4467 = The conversion value obtained from the photosynthetic equation (Baharuddin et al., 2014).

Conversion of Carbon Biomass to Oxygen (O_2)

1. **CO_2- equivalent**

   The calculation uses a ratio of the relative atomic mass of C (12) and the passive molecular mass of CO_2 (44) with the formula (Kemenhut,2013) :

$$CO_2 - ekuivalen = (\frac{44}{12}) \times C$$

2. **Oxygen Production (O_2)**

   Oxygen production/release is calculated based on the amount of oxygen produced during photosynthesis minus the amount used during plant respiration. The amount of oxygen production can be estimated from carbon sequestration based on atomic weight (Ross and Salisbury, 1978):

$$Net\ O^2 = net\ C \times \frac{32}{12}$$

The determination of carbon storage measurement plots is carried out with the following sizes: Plot size: "20m × 20m."

Sampling using purposive sampling and sampling methods without harvesting or logging. This sampling method involves measuring the height of oil palms and tree diameters by applying allometric equations to extrapolate biomass. Determination of the number of tiles that the measurement takes, using 10% IS. Based on Boon and Tideman (1950) for forests whose area ≥ 1,000 ha IS used 2%, while the area < 1,000 ha IS used 5% to 10%. Based on the area in the Arboretum of PT. Arutmin NPLCt is 5.05 ha.

The parameters measured are (dbh ≥ 10 cm and ≥ height 1.5 m) by measuring the diameter of the tree stand. Form a stand plot like Figure 1.

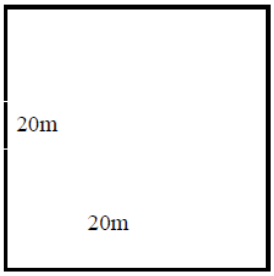

Figure 1. Plot Size Shape

RESULTS

Based on stand diameter measurements that have been carried out in the arboretum of PT. Arutmin Indonesia-NPLCT. The results obtained are biomass average 211.351 tons/ha, average carbon 99.33 tons/ha, carbon absorption 309.98, average CO2-equivalent 364.22, and average oxygen production 264.894 tons/ha. The results of these results can be seen in Table 2.

Table 2. Total Biomass, Carbon Stocks, Carbon Sequestration, CO2-equivalent, and Oxygen Production in Arboretums

No.Plot	Biomass Total (Ton/Ha)	Carbon Total (Ton/Ha)	Sequestration Carbon (Ton/Ha)	CO2-Ekuivalen (Ton/Ha)	Production Oksigen (Ton/Ha)
1	270,75	127,25	397,10	466,59	339,33
2	11,58	5,44	16,98	19,95	14,51
3	93,50	43,95	137,14	161,14	117,19
4	153,72	72,25	225,46	264,91	192,66
5	277,86	130,59	407,53	478,84	348,25
6	28,42	13,36	41,68	48,97	35,62
7	307,72	144,63	451,33	530,30	385,67
8	141,64	66,57	207,74	244,09	177,52
9	314,84	147,98	461,78	542,58	394,60
10	594,16	279,26	871,46	1023,94	744,68
11	153,88	72,32	225,69	265,18	192,86
12	188,16	88,43	275,97	324,26	235,83
Average	**211,351**	**99,335**	**309,989**	**364,229**	**264,894**
TOTAL	**2536,22**	**1192,02**	**3719,87**	**4370,75**	**3178,73**

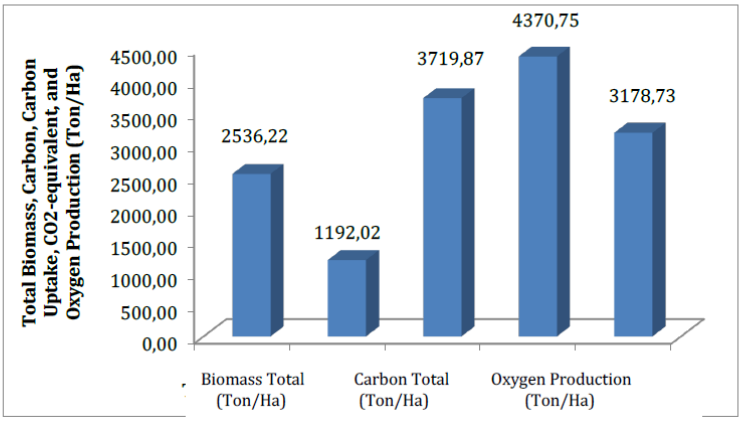

Figure 2. Diagram of total biomass, carbon, carbon sequestration, CO2-equivalent, and oxygen production (ton/ha)

Arboretum located in the area of PT. Arutmin Indonesia-NPLCT covering an area of 5.05 ha. The results obtained for total biomass, total carbon, total uptake, CO2-equivalent and total oxygen production in the area can be seen in table 3.

Arboretum located in the area of PT. Arutmin Indonesia-NPLCT covering an area of 5.05 ha. The results obtained for total biomass, total carbon, total uptake, CO2-equivalent and total oxygen production in the area can be seen in table 3.

Table 3. Total Biomass, Carbon Stocks, Carbon Sequestration, CO2-Equivalent, and Oxygen Production in an Arboretum covering an area of 5.05 ha

No	Arboretum Area (Ha)	Average Biomass (Ton/Ha)	Average Carbon (ton/ha)	Average Carbon Sequestration (ton/ha)	Average CO2-Ekuivalen (ton/ha)	Average Oxygen Production (ton/ha)
1	5.05	211,35	99,34	309,99	364,23	264,89
		Biomass Total (Ton)	Carbon Total (ton)	Carbon Sequestration (Ton)	CO2- Ekuivalen (Ton)	Oxygen Production (Ton)
2		1.067,32	501,64	1.565,45	1.839,36	1.337,71

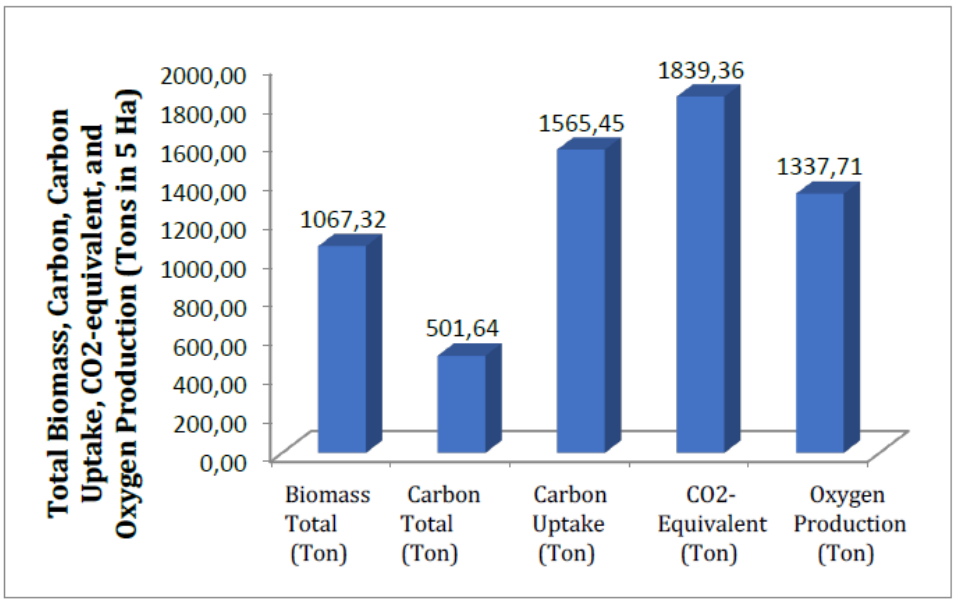

Figure 3. Diagram of total biomass, carbon, carbon sequestration, CO2-equivalent, and oxygen production in an area of 5.05 ha (tons)

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Jurnal change in greenes level

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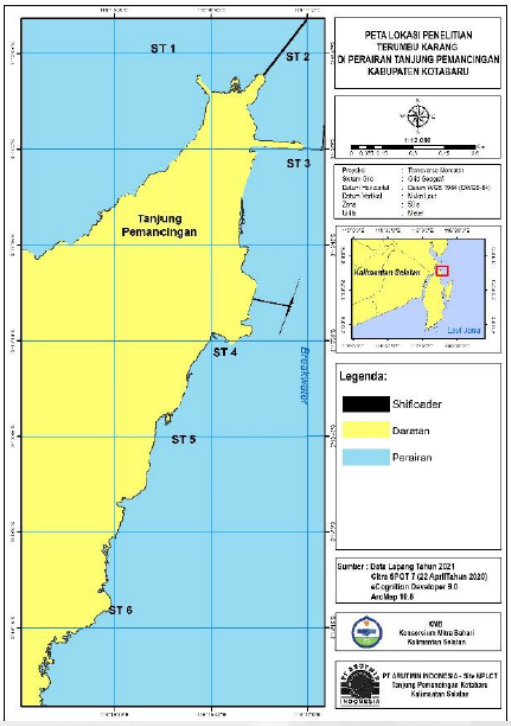
Mapping Of Subtrate Categories With The OBIA Method Using SPOT-7 Imagery On Fringing Reef At Tanjung Pemancingan, Kotabaru Regency

Hamdani[1,2], Melkyanus[2], Ananda Fitrian Anshary[2] Risman[2],
Arif Mahdian Noor[2] Yogi Swara Putra Mendra[3], Muhammad Subli[4]
[1]*Konsorsium Mitra Bahari Kalimantan Selatan, Indonesia*
[2]*Marine Science Department Lambung Mangkurat University*
[3]*PT. Arutmin Indonesia NPLCT - Tanjung Pemancingan, Kotabaru*
[4] *Kelompok Sahabat Laut Desa Sarang Tiung, Kotabaru*

II. Materials And Methods

Data collection activities were carried out during September and November 2021. This activity is located around the port operation area and coal terminal of PT Arutmin Indonesia NPLCT. The research area covers the fringing reef area along the east side to the north of the Kotabaru Tanjung Pemancingan. Starting from the flat fringing reef around Bangko Sarang Tiung bay to the north of the NPLCT loading jetty and its west side. A map of data collection locations can be seen in Figure 1.

The tools and materials used in this study include diving equipment, slates and pencils, underwater cameras, GPS (Global Positioning System, roll meters, 1.5 x 1.5 m plots, laptops, Microsoft Office 2019, ArcGIS Desktop 10.8, ENVI 5.3, ER Mapper 2015, eCognition Developer 64 bit version 9.1 and Surfer 1, SPOT-7 2020 Imagery.



Data Analysis

Initial Processing of Satellite Imagery

The initial processing stage of satellite imagery (image preprocessing) is carried out to improve the original image data (raw data) into satellite images ready for interpretation. The work carried out includes correcting errors due to the scattering of particles in the atmosphere recorded by satellite imagery (radiometric correction), repairing errors in the position of satellite image recording against earth references (geometric correction), masking and cropping, image composites and object sharpening in images through spectral value stretching. Image. Furthermore, water column correction is carried out to improve image quality by reducing interference in the water column.

There are two ways of correcting the water column, especially in finding the value (ki/kj), representing the attenuation coefficient ratio. First, through the gradient values on the linear lines formed by a pair of visible spectrum bands. Second, it can also be obtained based on the Lyzenga formula (1981) as follows:

$$Y = (\text{In } band\ 1) + \left(\frac{ki}{kj} \times \text{In } band\ 2\right)$$

$$\frac{ki}{kj} = \alpha + \sqrt{\alpha^2 + 1}$$

$$\alpha = \frac{(Varian\ band\ 1\text{-}Varian\ band\ 2)}{(2 \times covariant\ band\ 1\ dan\ band\ 2)}$$

$$K = \exp\left(-\frac{\|x - x'\|^2}{2\sigma^2}\right) \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

Where: X represents the vector of each data, represents the number of degrees of the polynomial function.

Validation/Accuracy Test

The accuracy test carried out in this study was to compare the classification results of benthic distribution maps with field data. The method used in this accuracy test is a confusion matrix based on the following equation (Congalton and Green, 2008 in Manan, 2019):

Overall Accuracy (OA) : ..... (2)

$$OA = \frac{\sum_{i}^{k=1} n_{ii}}{N}$$

Producer Accuracy (PA) : ..... (3)

$$PA = \frac{n_{ii}}{n_{i+}}$$

User Accuracy (UA) : ..... (4)

$$UA = \frac{n_{ii}}{n_{+i}}$$

Description:

N : the number of objects contained in the classification and reference results

n_{i+} : the total number of objects in the i-th row

n_{+i} : the total number of objects in the ith column

n_{ii} : the diagonal value of the matrix of the i-th row and i-column

The acceptable accuracy limit for shallow water bottom habitat maps based on SNI 7716:2011 concerning Mapping shallow sea bottom habitat is 60%.

III. Result

Water Column Correction

The water column correction is used by calculating DII (Deep Invariant Index) from the Lyzenga formula which uses information from the Ratio of the attenuation coefficient for each pair of visible light bands. The results of the calculation of the attenuation coefficient ratio of the visible light band pair of SPOT 7 images are presented in the following table:

Table 1. The Ratio of the attenuation coefficient of the visible light band pairs of the SPOT 7 image

Band Combination	Attenuation Coefficient (ki/KJ)
DII 2/3 (blue - green)	0.010752253
DII 2/4 (blue-red)	0.001771271
DII 3/4 (green - red)	0.003958538

Creating a Classification Scheme

The naming and description of the ten substrate category components are based on the dominant coverage of the substrate category at each observation point, composed of one or more substrate category components. The naming, description and code for each class of substrate category are presented in Table 2.

Table 2. Name and Code of Substrate Category

Substrate category class	Class Code	Description
Hard Corals	HC	*Hard corals* predominate
Hard Coral Mix	HC Mix	Mixed *hard corals*
Dead Coral with Algae	DCA	Dominant *Dead Coral with Algae*
Dead Coral with Algae Mix	DCA Mix	*Dead Coral with Algae* mixed
Dead Coral	DC	Dominant *Dead Coral*
Sand	S	*Sand* dominant
Sponges	SP	*Sponge* dominant
Hard Ground	HG	*Hard Ground* dominant
Soft Corals	SC	Dominant *Soft Coral*
Rock – Sand	RCK S	*Rock* mix *Sand*

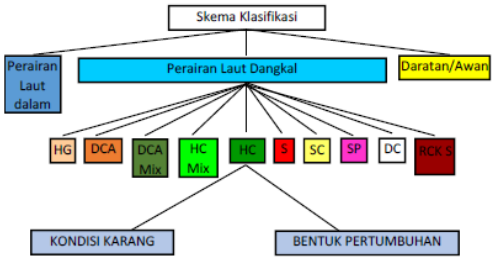
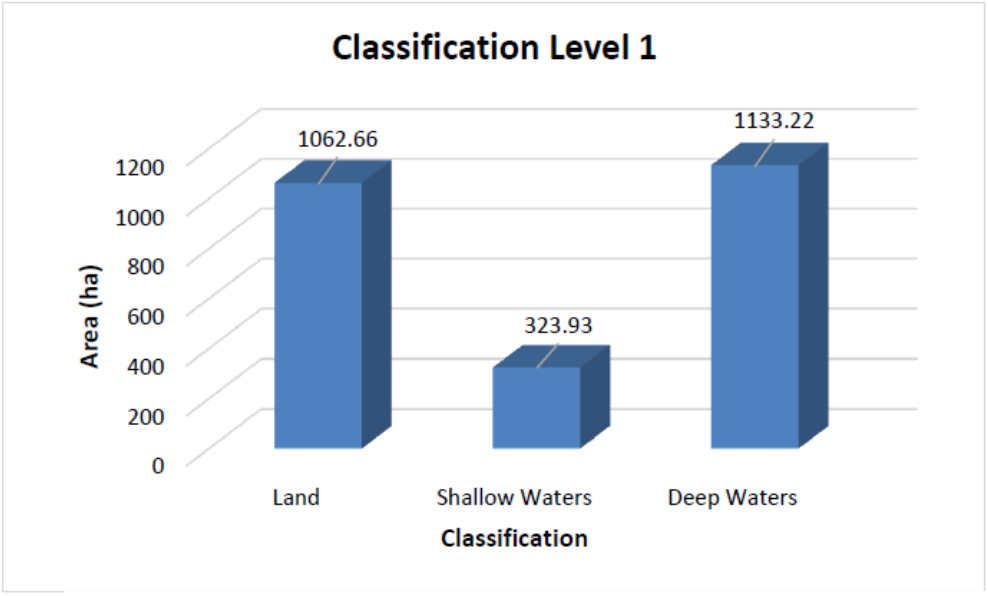


Figure 3. The results of the 2-level classification scheme

The classification scheme for shallow sea water substrate categories resulting from 250 observation points will then be divided into two, namely, 185 observation points will be used as RoI (region of interest) data in the image classification process. Then the remaining 65 observation points will be used as data for testing the accuracy of the image classification results. Classification schemes for the ten resulting shallow water substrate categories will all be used in the SPOT 7 image classification process.



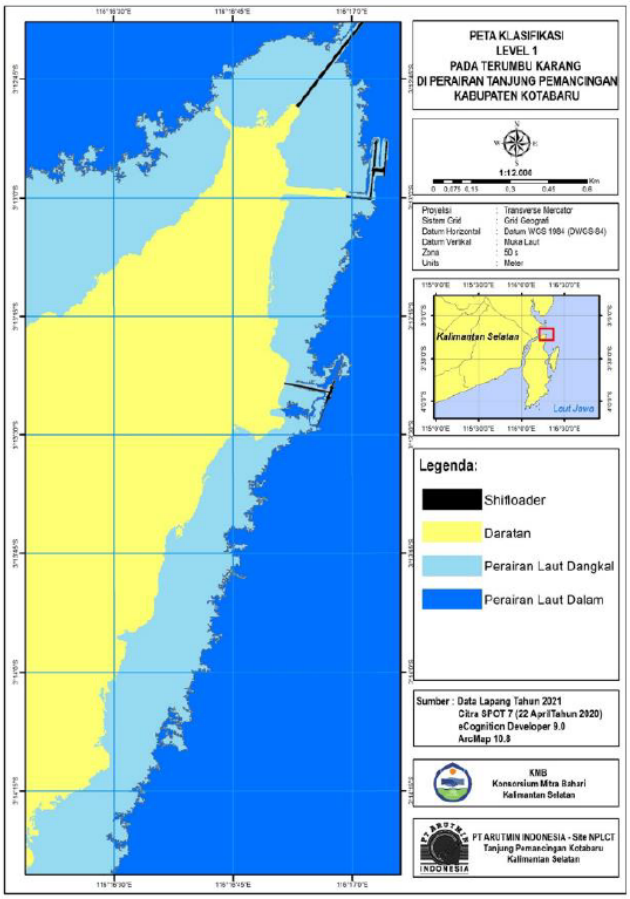


Figure 5. Class Area of level 1 Classification Results

Table 3. Producer Accuracy, User Accuracy and Overall Accuracy for Ten Benthic Categories

Field / Image	HG	HC	DCA mix	SP	SC	DCA	S	HC Mix	DC	Sum	User Accuracy
HG	6	0	0	0	0	0	4	0	0	10	60
HC	0	31	2	4	3	3	2	1	0	46	67,39
DCA mix	1	0	4	0	0	1	0	0	0	6	66,67
SP	0	0	0	2	0	0	0	0	0	2	100
SC	0	0	0	0	10	0	0	0	0	10	100
DCA	0	2	0	0	0	5	3	0	0	10	50
S	1	1	1	0	0	1	4	2	1	11	36,36
HC mix	0	1	0	0	0	0	1	3	0	5	60
DC	0	0	0	0	0	0	0	0	1	1	100
Sum	8	35	7	6	13	10	14	6	2	101	
Producer Accuracy	75	88.57	57,14	33,33	76,92	50	28.57	50	50	OA	65,35

The results of the classification of substrate categories (level 2) using the SVM algorithm are presented in Figure 6 for the application of the ten classes of substrate categories and Figure 7 for the percentage diagram of the area covered by the substrate categories in the ten classes of substrate categories, presented as follows:

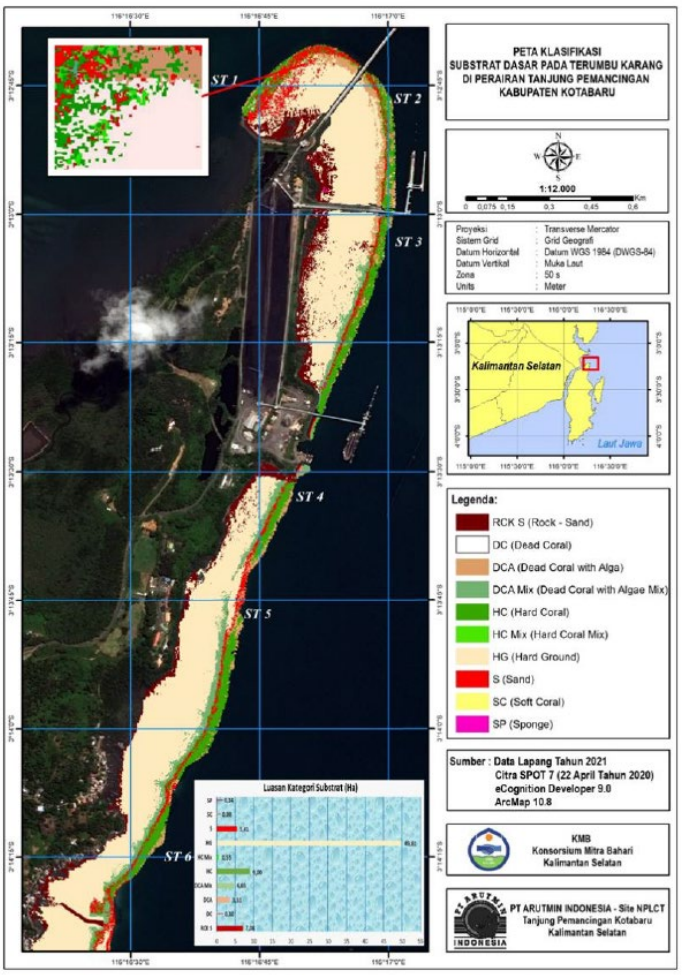


Figure 5. Classification Map of 10 Classes of Substrate Categories on Fringing Reefs at Tanjung Pemancingan Using a Scale of 3 on SPOT 7 Imagery

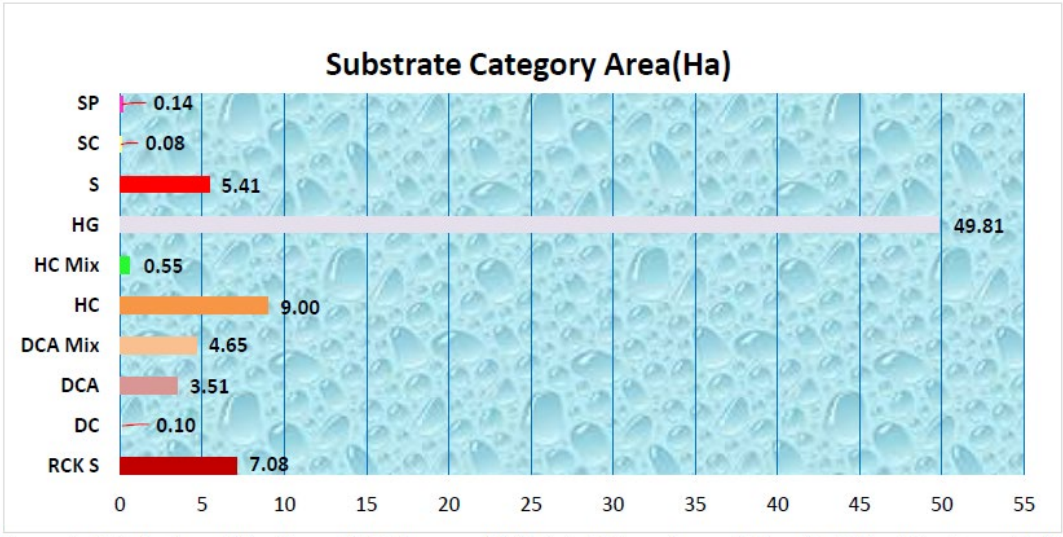


Figure 6. Distribution of the Area of 10 Classes of Substrate Categories as A Result of Classification with The SVM Algorithm on The Segmentation Scale 3 SPOT Image 7

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Changes In Greenness Level In PT Arumin Indonesia's RPT NPLCT Area With Normalized Difference Vegetation Index (NDVI)

Ibnu Dwi Thousand[1], Dimas Indrayanto Putra[2], Rika Putri Ardianti[3]
Universitas Lambung Mangkurat, Indonesia[1]
UPN Veteran Yogyakarta, Indonesia[2]
Universitas Lambung Mangkurat, Indonesia[3]
Email: ibnudwithousand002@gmail.com, dimasindrayanto@gmail.com, rikaputri0903@gmail.com

METHODS

This analysis was conducted at NPLCT PT Arutmin Indonesia, Pulau Laut Utara District, Kotabaru Regency, South Kalimantan. The analysis is carried out during 3 months from preparation, collection and processing.

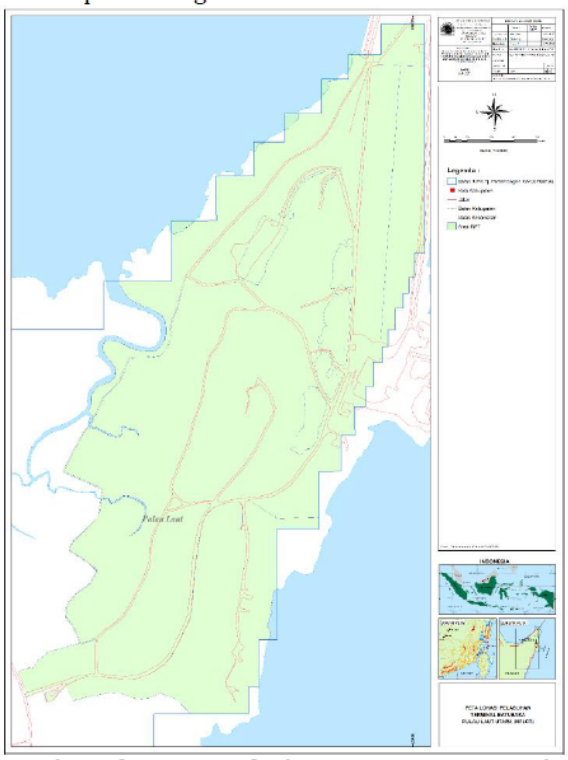


Figure 1. Location of *NDVI Analysis* at NPLCT PT Arutmin Indonesia

The object of this research is changes in the level of greenness with the *NDVI method* at NPLCT PT Arutmin Indonesia.

3. Data Analysis
   a. Changes in the greenish index in the observation area using the *Normalized Difference Vegetation Index* (NDVI) method which is based on a combination of pixel values of infrared band images or NIR and red bands. This method produces 7 classes of greenish level changes, namely land is not vegetated, low vegetation, rather low vegetation, medium vegetation, high enough vegetation, high vegetation and very high vegetation. In these 7 classes of greenness level change, there will be areas that experience reforestation, deforestation and do not change (Bashit, 2019).
   b. This greenness change refers to the Circular of the Chief Inspector of Mines (KAIT) No.9E/MB.07/DBT.PL/2023 concerning the spatial database of planning, implementation, and reporting of reclamation and post-mining in mineral and coal mining business activities.

Table 1. *NDVI value*

Class	Color	NDVI	Information
1	Blue	-1 to -0.1	Non-vegetated
2	Red	0.1 to 0.3	Low vegetation
3	Oren	0.3 to 0.5	Vegetation is rather low
4	Yellow	0.5 to 0.6	Medium vegetation
5	Light Green	0.6 to 0.7	Vegetation is quite high
6	Toska Green	0.7 to 0.9	High Vegetation
7	Dark Green	0.9 to 1	Very High Vegetation

Figure 2. Research Location Benchmarking

After cutting the image, radiometric correction is performed. This radiometric correction is to reduce noise such as lighting, color and clouds contained in the location imagery. All five images are 100% cloud-clean, so no cloud radiometric correction is required on the imagery. Satellite resources are said to be good if the area of cloud coverage is less than 10% (Danoedoro et al., 2019).

Sentinel-2A Image Analysis

The *Normalized Difference Value Index* (NDVI) analysis method calculates the pixel value by combining the infrared band or NIR and the red band. The band combination continued to divide the greenish level into 7 classes, namely unvegetated land, very low vegetation, low vegetation, medium vegetation and high vegetation (Faizal et al., 2023). After the division of 7 classes of greenish level, the results of greenness changes were obtained by interpolation from 2019 to 2023. The results of this interpolation get 3 changes, namely reforestation, deforestation and no change.

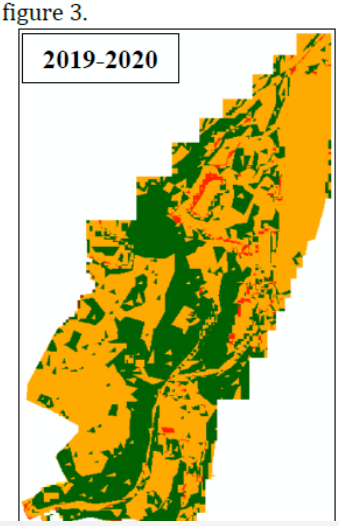
Table 2. Greenness Level 2019 to 2023

Information	2019	2020	2021	2022	2023
Non-vegetated	15,35	14,52	12,34	14,08	15,91
Low vegetation	12,81	7,17	4,27	8,00	27,29
Vegetation is rather low	67,30	43,64	7,20	55,19	55,28
Medium vegetation	2,95	31,31	6,07	20,99	0
Vegetation High enough	0	1,76	14,47	0,15	0
High Vegetation	0	0	54,09	0	0
Very High Vegetation	0	0	0	0	0
Total	98	98	98	98	98

Table 3. Greenish Rate Interpolation From 2019 to 2023

Information	Year				
	2019-2020	2020-2021	2021-2022	2022-2023	2019-2023
Reforestation	38,41	79,09	1,08	0,47	2,27
Deforestation	1,65	1,35	79,48	40,65	20,35
Remain	58,14	17,76	17,68	57,17	75,64
Total	98	98	98	98	98

The results of *the NDVI* image overlay from 2019 to 2020 obtained an area of land that experienced reforestation or an increase in greenness level of 38.41 ha. Deforestation or decrease in greenness levels that occurred covering an area of 1.65 ha. The area that has not changed is 58.14 ha. Seen in figure 3.



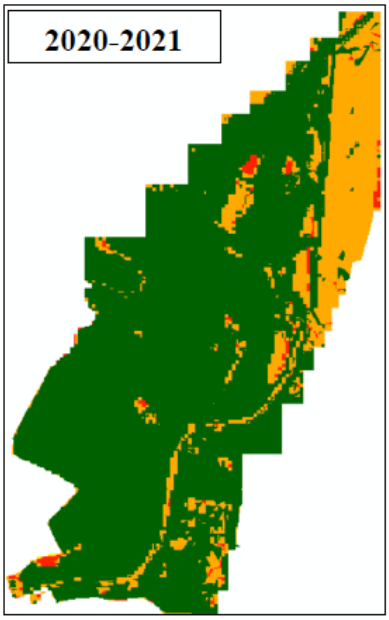


Figure 4. Greenish Rate Interpolation from 2020 to 2021

The results of *the NDVI* image overlay from 2021 to 2022 obtained an area of land that experienced reforestation or an increase in greenness level of 1.08 ha. Deforestation or decrease in greenness levels that occurred covering an area of 79.48 ha. The area that has not changed is 17.68 ha. Seen in figure 5.

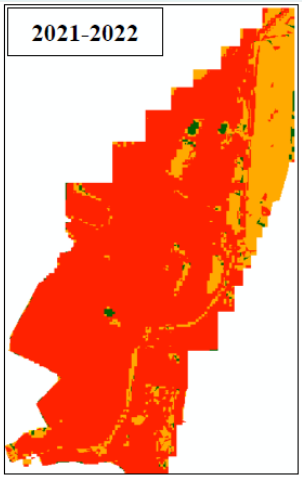


Figure 5. Greenish Rate Interpolation from 2021 to 2022

The results of the *NDVI* image overlay from 2022 to 2023 get an area of land that has undergone reforestation or an increase in greenness level of 0.47 ha. Deforestation or decrease in greenness levels that occurred covering an area of 40.65 ha. The area that has not changed is 57.17 ha. Seen in figure 6.

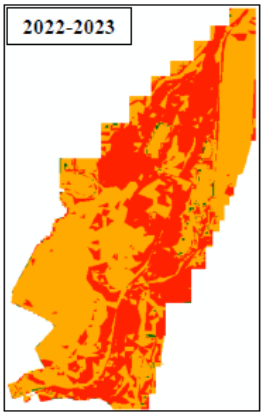


Figure 6. Greenish Rate Interpolation from 2022 to 2023

The results of *the NDVI* image overlay from 2019 to 2023 get an area of land that has undergone reforestation or an increase in greenness level of 2.27 ha. Deforestation or decrease in greenness levels that occurred covering an area of 20.35 ha. The area that has not changed is 75.64 ha. Seen in figure 7.

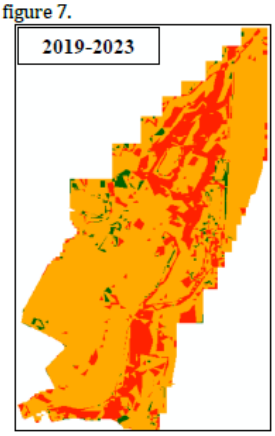


Figure 7. Greenish Rate Interpolation from 2019 to 2023

The graph of change shows that the largest area of 79.09 ha that experienced an increase in greenness occurred between 2020 and 2021. The rapid change in deforestation or decrease in greenness occurred between 2021 and 2022 with an area of 79.48 ha.

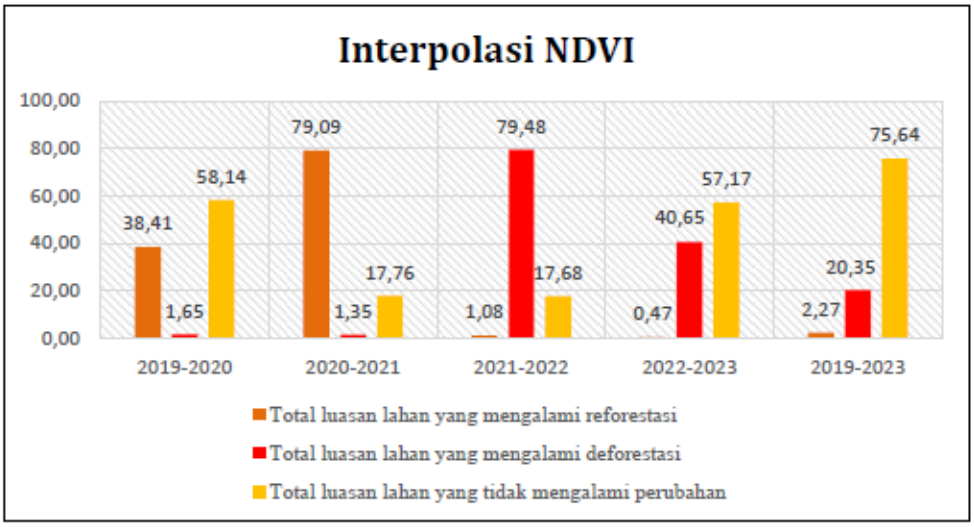


Figure 8. Interpolation graph of *NDVI changes* from 2019 to 2023

CONCLUSION

Mangrove vegetation health index analysis by implementing NDVI (normalized difference vegetation index) classification method on sentinel-2 image data case study: Segara Anakan, Kabupaten Cilacap. *IOP Conference Series: Earth and Environmental Science*, *584*(1), 12069. IOP Publishing.

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IV. Conclusion

Analysis of SPOT 7 satellite imagery using the OBIA (object base image analysis) classification method with the SVM (support vector machine) algorithm shows that the classification of 10 categories of substrate classes having an OA (overall accuracy) of 65.35% already fulfils the minimum SNI accuracy test criteria of 60%. The calcification map of 10 substrate categories shows that the biotic category HC (hard coral), which is coral has an area of about 9 Ha, the category of coral mixed with other substrates (HC Mix) is around 0.55 Ha, the substrate category SP (sponge) is around 0.17 Ha and SC (soft coral) around 0.10 Ha. Furthermore, DCA mix or dead coral covered by algae and mixed with other substrates with other substrate categories around 4.65 Ha, followed by DCA around 3.51 Ha and DC around 0.1 Ha. While abiotic elements such as sand (S) are around 5.41 Ha, then hard ground (HG), the flat reef that is dead and covered with sediment, has an area of around 50 Ha. Another abiotic category, RCK-S covering an area of 7.06 Ha, is a category of andesite rock and sand substrates near coral reefs and beaches.

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Miranda Verly, O., Vieira Leite, R., da Silva Tavares-Junior, I., José Silva Soares da Rocha, S., Garcia Leite, H., Marinaldo Gleriani, J., Paula Miranda Xavier Rufino, M., de Fatima Silva, V., Moreira